

Mignot Mesele

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EDUCATION

University of California, Santa Cruz

Santa Cruz, CA

Bachelor of Science in Computer Science, GPA: 3.58

June 2025

Master of Science in Computer Science and Engineering

June 2027

Relevant Coursework: Software Design, **Analysis of Algorithms**, **Machine Learning**, **Data Structures and Algorithms**

SKILLS

Programming Languages: Python, Java, JavaScript, HTML/CSS, C, C++, SQL, Kotlin, Go

Tools and Frameworks: React, Next.js, Node.js, Django, Flask, PostgreSQL, Stripe, AWS, Docker, Git/Github, Selenium

WORK EXPERIENCE

UCSC PD

Jan 2025 - June 2025

Software Engineering Intern

hybrid

- **Developed and deployed** a production-ready crime visualization dashboard using **Power BI**, **Node.js**, **JavaScript**, **Docker**, and **PostgreSQL**, delivering real-time insights that improved accessibility of campus crime data for 20,000+ stakeholders.
- Centralized reporting access for students, police, and community members via a user-friendly platform.
- Automated data pipelines and ensured scalability through containerized deployment.

GlobiFYE

Sep 2024 - Apr 2025

Software Development Intern

Remote

- Collaborated in an agile team to build a scalable job-listing and networking platform, using **React**, **Node.js**, **Express.js**, **PostgreSQL**, and **Azure**.
- Implemented blog features, enabling users to share professional content and event updates.

RESEARCH EXPERIENCE

UCSC

July 2025 - Present

Graduate Student Researcher

Santa Cruz, CA

- **Integrating reinforcement learning methods** to optimize quantum processing unit (QPU) placement, benchmarking RL-based strategies against heuristic baselines to enhance job scheduling efficiency in a quantum cloud simulator.

PROJECTS

TCG Tracker

- Collaborated with a team to create a trading card game collection application that enables users to grade and calculate the value of their cards using **Javascript**, **Vite**, **React**, **Firebase**, **Google Cloud Vision**, and **Node.js**.

Zema

- Created a **full stack** web application that empowers users to track and analyze eczema flare-ups by integrating **AI-powered** image scanning using **Javascript**, **Vite**, **React**, **Flask**, **Firebase**, **Tensorflow**, **Python**, and **TailwindCSS**.

Chat ai

- Built a streamlined application using **Next.js**, **React**, **Material-UI**, **Firebase**, and **vercel**, featuring a chat interface for user interaction with the **Llama 3.1 LLM** via the **Groq API**, providing a chatbot that users can converse with.

Cardify

- Developed a **full stack AI Flashcard SaaS** application using **Next.js**, **React**, **Clerk**, **Firestore**, **OpenRouter API**, **Vercel**, **Stripe** to provide users with easy access to dynamic flashcard creation and management.

PUBLICATION

- Zhou, R., Gan, Y., Shen, L., Liu, Y., Mesele, M., & Qian, C. (2025). QuCloudSim: A Customizable Discrete Event Simulator for Quantum Cloud Computing Environment. IEEE QCNC 2025.